Random Signs into Matchings

A Godsil–Gutman identity, formalized in Lean 4

Carles Marín*

Independent researcher

karlesmarin@gmail.com

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Abstract

I report a machine-checked formalization, in Lean 4 / Mathlib, of the Godsil–Gutman identity: the average characteristic polynomial of a uniformly random ±1 signing of a graph is its *matching polynomial*. Along the way I formalize the matching polynomial itself and its deletion recurrence—to my knowledge, the first such development in any proof assistant. I then isolate the one ingredient that makes the proof work, a $\mathbb{Z}/2$ *sign-averaging* fact ($\sum_{s=\pm1} s^k = 2\,[k \text{ even}]$), and show that the *same* formal lemma underlies a second, moment-level identity (the parity-closed-walk count of Chen–van Dam–Bu). Finally I formalize the Bilu–Linial spectral decomposition of a signed 2-lift, the bridge that connects all of this to Ramanujan graphs. I am deliberately precise about what is and is not done: the headline theorems are `sorry`-free and depend only on the three standard axioms, but the deep half of the Marcus–Spielman–Srivastava program (Heilmann–Lieb real-rootedness, interlacing families) is mapped, not formalized. This is a pearl, not a landmark; I hope it is a useful first stone.

1 A question about signings

Take a finite graph G and put a sign, $+1$ or -1, on each edge. The resulting *signed adjacency matrix* A_s has a characteristic polynomial $\det(xI - A_s)$. Different signings give different polynomials. A natural question—and the starting point of a beautiful chapter of spectral graph theory—is:

> *what do you get if you average* $\det(xI - A_s)$ *over all* $2^{|E|}$ *signings?*

The answer, due to Godsil and Gutman [3], is striking: the average is exactly the *matching polynomial* $\mu_G(x) = \sum_k (-1)^k m_k\, x^{n-2k}$, where m_k counts the k-edge matchings of G (Figure 1). A spectral average over random signs equals a purely combinatorial count of matchings.

Why should anyone outside spectral graph theory care, and why formalize it? Two reasons travel together through this paper.

It is the door to Ramanujan graphs. A signing of G is the same data as a 2-lift (double cover) of G, and the "new" eigenvalues of the lift are precisely the eigenvalues of A_s (Bilu–Linial; Section 5). So the average characteristic polynomial of the new eigenvalues is μ_G. Heilmann and Lieb [5] proved that the roots of μ_G all lie in $[-2\sqrt{\Delta-1}, 2\sqrt{\Delta-1}]$—the Ramanujan threshold

*The Lean formalization was carried out with Claude (Anthropic) as an AI research instrument; all design decisions, mathematics, and claims are the author's responsibility.

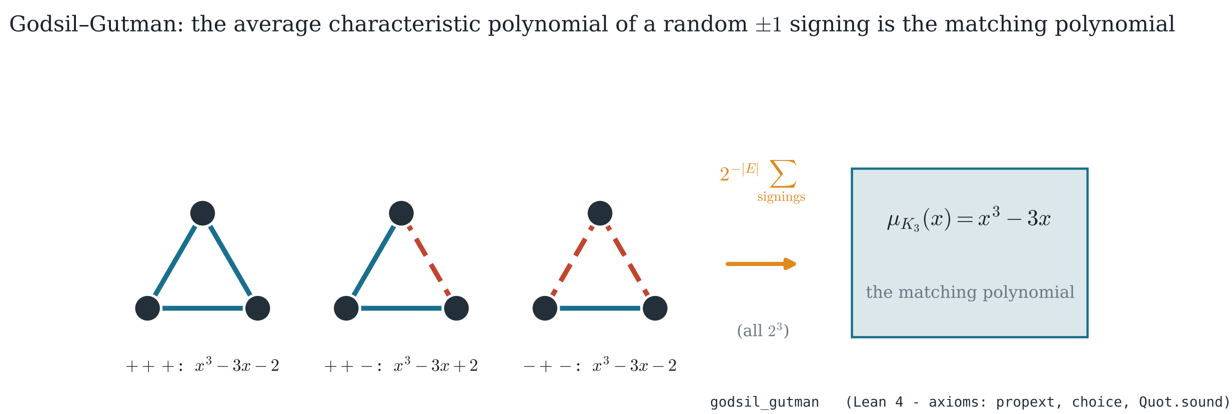

Figure 1: Godsil–Gutman on K_3: each ± 1 signing (teal $+$, dashed red $-$) gives a characteristic polynomial; their average is the matching polynomial $\mu_{K_3}(x) = x^3 - 3x$. Machine-checked as `godsil_gutman`.

(Figure 2). Marcus, Spielman and Srivastava [6] then used an interlacing-families argument to show that *some* signing achieves a lift whose new eigenvalues are all within that threshold, proving the existence of Ramanujan graphs of every degree. Godsil–Gutman is the first link of that chain.

It hides a single, reusable pattern. The proof of Godsil–Gutman, when one looks closely, rests on one tiny fact: averaging a sign monomial over ± 1 kills every odd power and doubles every even one. This is a $\mathbb{Z}/2$ Fourier projection, and—this is the part I found worth telling—the *same* formal lemma drives an apparently different, moment-level identity (Section 4). Formalization makes such shared structure not a slogan but something a reader can check by name.

Contributions. I contribute, all in Lean 4 / Mathlib and all `sorry`-free with `#print axioms` reporting only `propext`, `Classical.choice`, `Quot.sound` (Table 1):

1. the matching polynomial, the matching number, and the deletion recurrence $m_{k+1}(G) = m_{k+1}(G - v) + \sum_{u \sim v} m_k(G - v - u)$;
2. a complete proof of the Godsil–Gutman identity (`godsil_gutman`);
3. the $\mathbb{Z}/2$ sign-averaging engine and its two consumers—Godsil–Gutman at the characteristic-polynomial level and the parity-closed-walk kernel at the moment level;
4. the Bilu–Linial 2-lift charpoly factorization (`charpoly_twoLift`).

I am equally explicit about the boundary of the work (Section 6): Heilmann–Lieb and the interlacing-families step are mapped in detail but not formalized.

2 What exactly is being averaged?

Before a machine can check anything, the objects must be pinned down. I work over an arbitrary finite vertex type V.

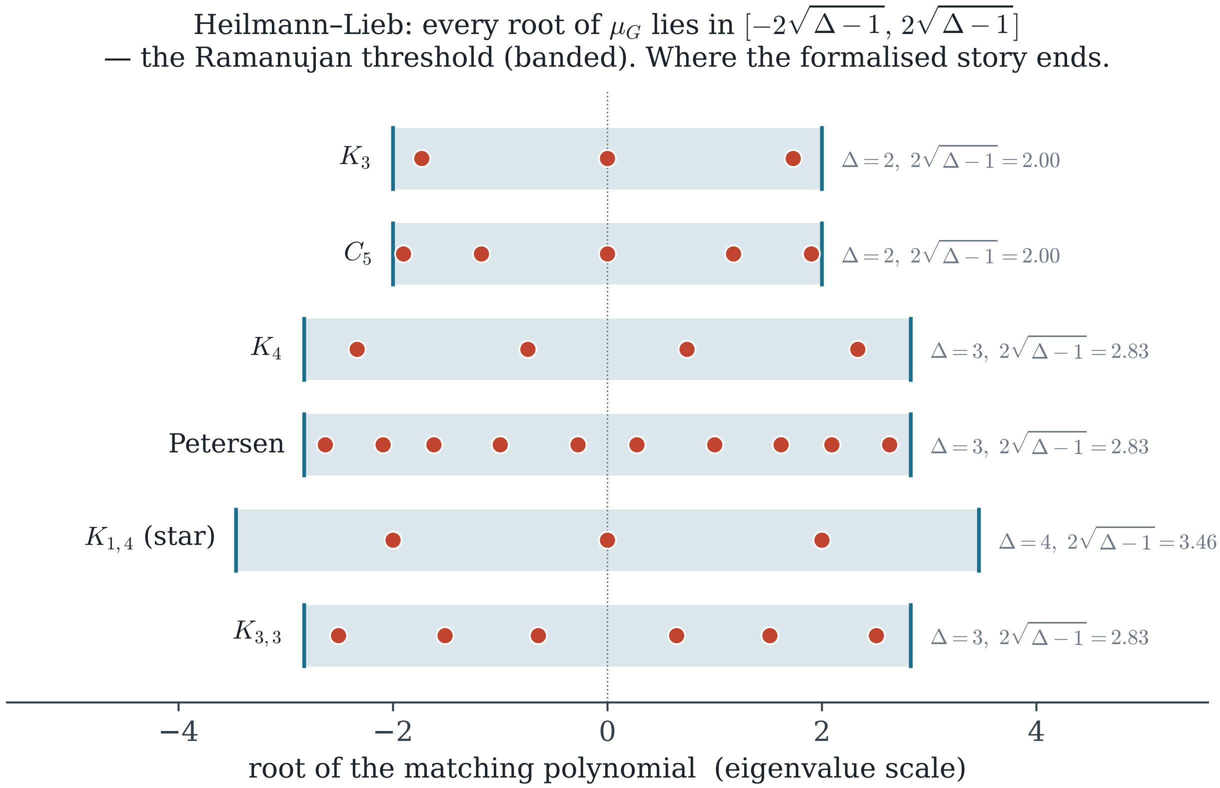

Figure 2: Heilmann–Lieb: the (computed) roots of the matching polynomial all lie inside the band $[-2\sqrt{\Delta-1},\, 2\sqrt{\Delta-1}]$, the Ramanujan threshold. This bound is the endpoint of the program; here it is mapped (Section 6), not yet formalized.

Definition 1 (Signed adjacency)**.** *A signing is a function* $\mathrm{cfg} \colon \mathrm{Sym2}\, V \to \mathrm{Bool}$ *on unordered pairs. The signed adjacency matrix* A_{cfg} *is the symmetric real matrix that is* ± 1 *on edges (according to* cfg*) and* 0 *elsewhere.*

A small but honest subtlety surfaces immediately, and it is worth stating plainly rather than hiding it. I let cfg range over *all* unordered pairs, not only edges; the non-edge bits are simply ignored by A_{cfg}. Consequently each genuine signing is counted $2^{|\mathrm{Sym2}\, V| - |E|}$ times, and my theorem reads

$$\boxed{\sum_{\mathrm{cfg}} \det(xI - A_{\mathrm{cfg}}) \;=\; (\#\mathrm{cfg}) \cdot \mu_G}\,, \qquad \#\mathrm{cfg} = 2^{|\mathrm{Sym2}\, V|}.$$

Dividing by #cfg recovers the textbook edge-average $2^{-|E|} \sum_{\mathrm{signings}} = \mu_G$ exactly; the redundant non-edge bits cancel. I chose the all-pairs phrasing because it makes the proof's $\mathbb{Z}/2$-Fourier structure (Section 4) uniform over the whole index type, at the cost of this one bookkeeping remark.

Definition 2 (Matching polynomial)**.** $\mu_G(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k m_k(G)\, x^{n-2k}$, *where* $m_k(G)$ *is the number of* k*-subsets of edges that are pairwise vertex-disjoint and* $n = |V|$.

Mathlib has a rich theory of simple graphs and of characteristic polynomials, but no matching polynomial; building it (and proving its deletion recurrence) was a prerequisite, and is, to my knowledge, the first such infrastructure in a proof assistant.

Table 1: The formalised results (Lean 4 / Mathlib). All theorems are `sorry`-free and non-vacuous; `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`.

Lean name	Statement	Status
`godsil_gutman`	$\sum_{\text{cfg}} \det(xI - A_{\text{cfg}}) = \#\text{cfg} \cdot \mu_G$	proven
`matchingPoly` (+infra)	$\mu_G = \sum_k (-1)^k m_k\, x^{n-2k}$ (first in any prover)	defined
`matchingNumber_recurrence`	$m_{k+1}(G) = m_{k+1}(G{-}v) + \sum_{u\sim v} m_k(G{-}v{-}u)$	proven
`charpoly_twoLift`	$\det\big(xI - \left[\begin{smallmatrix} B & C \\ C & B \end{smallmatrix}\right]\big) = \mu_{B+C}\, \mu_{B-C}$	proven
`signAvg_ne_zero_iff`	sign-monomial survives $\iff$ multiplicities even	proven
`sum_signOf_prod_pow`	$\mathbb{Z}/2$ parity-projection kernel (moment level)	proven
Heilmann–Lieb	$\text{roots}(\mu_G) \subseteq [-2\sqrt{\Delta-1},\, 2\sqrt{\Delta-1}]$	future
interlacing families	$\exists$ signing with $\lambda_{\max} \le \lambda_{\max}(\cdot)$	future

Table 2: Numerical cross-checks of the formalised identities (SymPy, exact). Left: the average characteristic polynomial of all ± 1 signings equals μ_G (Godsil–Gutman). Right: the average d-th spectral moment equals the number of parity-closed walks (here $d = 4$).

G	$2^{-\lvert E\rvert}\sum_{\text{signings}} \det(xI - A_s)$	$= \mu_G$?	$\overline{\text{tr}(A_s^4)} = P_4$
K_3	$x^3 - 3x$	✓	18
C_4	$x^4 - 4x^2 + 2$	✓	24
C_5	$x^5 - 5x^3 + 5x$	✓	30

3 How a machine proves Godsil–Gutman

The classical one-paragraph proof expands $\det(xI - A_{\text{cfg}})$ over permutations, sums over signings, and observes that only permutations that are *involutions made of edges*—that is, matchings—survive. Turning that paragraph into a checked proof meant making each "observe that" explicit. The architecture is a short pipeline; I sketch it and flag the two places that fought back.

Only edge-involutions survive. Writing the determinant as $\sum_\sigma \text{sign}(\sigma) \prod_i (\cdots)$ and summing over signings, the contribution of a permutation σ is, up to sign and a power of x, an inner sum over signings of a product along σ's support. I prove (`innerSum_eq_ite`) that this inner sum is #cfg when σ is an edge-involution ($\sigma^2 = 1$ with every 2-cycle an edge) and 0 otherwise. The "0 otherwise" is exactly the $\mathbb{Z}/2$ cancellation: a permutation with a non-paired edge contributes a sign that averages away.

Edge-involutions are matchings. The surviving permutations biject with matchings: an edge-involution σ maps to its set of 2-cycle edges, and a matching maps back to the involution that swaps each edge's endpoints. I build this bijection explicitly—`matchingOfPerm` one way, `permOfMatching` the other— and prove both round-trips. A pleasant surprise was that the inverse map is *proof-irrelevant* in the matching hypothesis (the permutation's data depends only on the matching, not on the proof that it is one), which let the `Finset.sum_bij'` obligations discharge cleanly.

The two hard spots. First, a coercion: casting $(-1)^k$ from the unit group $\mathbb{Z}^\times$ through $\mathbb{Z}$ into $\mathbb{R}[x]$. The natural Mathlib rewrite (`Units.val_pow_eq_pow_val`) is definitionally a no-op and silently fails to rewrite; I needed a short induction (`intCast_units_neg_one_pow`). Second, the final reassembly required a fibered sum over matching size with a boundary term, handled with

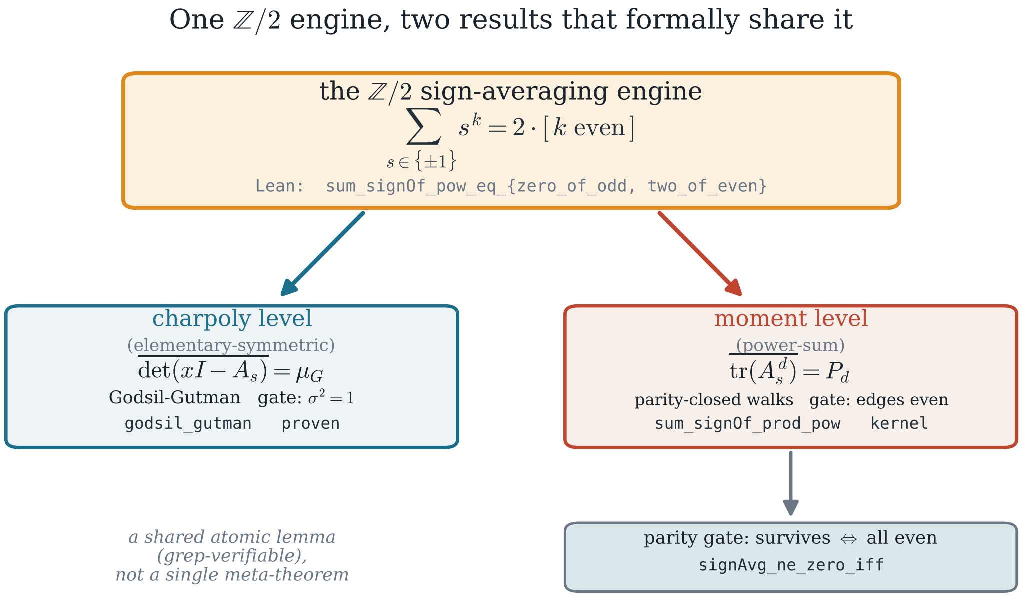

Figure 3: One $\mathbb{Z}/2$ sign-averaging fact (`sum_signOf_pow`) feeds both Godsil–Gutman (characteristic-polynomial level) and the parity-closed-walk kernel (moment level). The unification is a shared atomic lemma, visible in the dependency graph—not a single meta-theorem; the parity gate (`signAvg_ne_zero_iff`) is itself downstream of the moment kernel.

`Finset.sum_fiberwise_of_maps_to` and the matching-size bound $2k \le n$. Neither is deep, but both are the kind of friction a formalization must absorb and a paper should be honest about.

The result, `godsil_gutman`, is the precise statement of Section 2, machine-checked.

4 The engine underneath: one $\mathbb{Z}/2$ fact, two identities

Why does the cancellation in Section 3 happen? Because of a single arithmetic fact about signs:

$$\boxed{\sum_{s\in\{\pm1\}} s^k \;=\; \begin{cases} 2 & k \text{ even,} \\ 0 & k \text{ odd} \end{cases}}.$$

In Lean this is two one-line lemmas (`sum_signOf_pow_eq_two_of_even` and `_zero_of_odd`). Averaging a product of edge-signs over all signings factors, by `Fintype.prod_sum`, into a product of these per-edge sums; so *a sign monomial survives the average if and only if every edge multiplicity is even.* I package this as the parity gate (`signAvg_ne_zero_iff`; Figure 3).

Here is the part I find genuinely pleasant, and I state it carefully to avoid over-claiming. The very same per-edge lemmas drive a *second*, seemingly unrelated identity. Chen, van Dam and Bu [1] observe that the average d-th spectral moment of a signed graph counts its *parity-closed walks* (closed walks using each edge an even number of times). At the level of elementary symmetric functions one gets Godsil–Gutman; at the level of power sums one gets the parity-closed-walk count. They are two readings of the same $\mathbb{Z}/2$ projection. I formalize the algebraic kernel of the second identity (`sum_signOf_prod_pow`)—its full walk-count form needs walk infrastructure Mathlib lacks, which I do not claim.

A point of intellectual honesty. The unification here is real but modest, and I describe it exactly. I did *not* prove a single meta-theorem of which both results are corollaries. What is true—and formally checkable by name—is that both proofs invoke the *same* atomic lemmas `sum_signOf_pow_eq_*`. The shared structure lives in the dependency graph, not in a grand statement, and Figure 3 draws it that way.

5 The bridge to Ramanujan: 2-lifts

What does any of this have to do with expanders? The link is the 2-lift. A signing s of G determines a double cover G_s whose adjacency matrix has the block form $\left[\begin{smallmatrix} A_+ & A_- \\ A_- & A_+ \end{smallmatrix}\right]$, with A_+, A_- the positive- and negative-edge parts. I formalize the spectral decomposition of this block matrix (`charpoly_twoLift`):

$$\boxed{\det\!\left(xI - \begin{bmatrix} B & C \\ C & B \end{bmatrix}\right) \;=\; \det(xI - (B{+}C)) \cdot \det(xI - (B{-}C))},$$

proved by conjugating with the involution $\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ (which squares to $2I$) into block-diagonal form. With $B{+}C = A$ the ordinary adjacency and $B{-}C = A_s$ the signed one, this says the lift's spectrum is $\operatorname{spec}(A) \uplus \operatorname{spec}(A_s)$: the *new* eigenvalues are exactly those of the signed adjacency matrix. So Godsil–Gutman is the statement that their average characteristic polynomial is μ_G—and Heilmann–Lieb is the statement that μ_G's roots respect the Ramanujan threshold.

I am careful to say what I formalized: the *linear algebra* of the 2-lift (the matrix identity above), not the graph-theoretic 2-lift construction. That is the right building block, but it is a matrix theorem with a graph reading, and I present it as such.

6 How the story ends, and where I stopped

It would be dishonest to pretend I reached the end. I did not. But I can map it precisely, because the catalog of what remains is short and the route is classical.

- ✓ Godsil–Gutman (average charpoly $= \mu_G$).
- ✓ The 2-lift bridge (new eigenvalues = signed eigenvalues).
- ✓ The $\mathbb{Z}/2$ engine tying the spectral and combinatorial sides.
- ∘ **Heilmann–Lieb**: $\operatorname{roots}(\mu_G) \subseteq [-2\sqrt{\Delta-1}, 2\sqrt{\Delta-1}]$. The clean route is Godsil's path-tree: μ_G divides the matching polynomial of the path-tree $T(G,u)$; T is a forest, on which the matching polynomial *equals* the characteristic polynomial (signs are gauge-trivial), hence is real-rooted with the tree spectral-radius bound $2\sqrt{\Delta-1}$. This yields real-rootedness and the bound in one stroke, and needs no external interlacing machinery.
- ∘ **Interlacing families** [6]: the family $\{\det(xI - A_s)\}_s$ is interlacing, so some member's largest root is at most the largest root of the average μ_G, which is $\leq 2\sqrt{\Delta-1}$.
- ∘ Hence a Ramanujan 2-lift exists, and Ramanujan graphs of every degree follow by iterating.

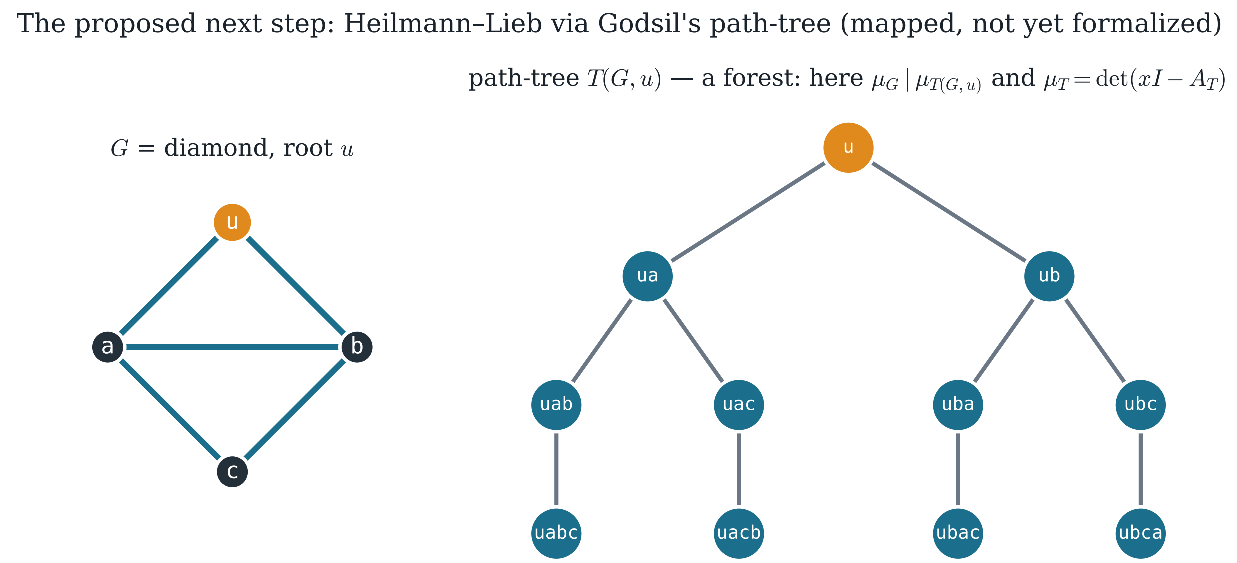

Figure 4: Godsil's path-tree $T(G, u)$ of the diamond G (root u, amber): its vertices are the paths of G starting at u, joined when one extends the other by an edge. $T(G, u)$ is a forest, so on it the matching polynomial *equals* the characteristic polynomial of the adjacency matrix; and μ_G divides $\mu_{T(G,u)}$. These two facts deliver Heilmann–Lieb. The construction is mapped here, not yet formalized.

Steps 4–6 are a multi-month undertaking; a literature check (Mathlib plus a web survey) found no prior formalization of Heilmann–Lieb or of the matching polynomial in any prover, so the road is unworn but also unbuilt. I think the honest contribution of this paper is to have laid, and verified, the first few paving stones, and to have drawn the rest of the map to scale.

7 The next stone: a path-tree proposal

The map in Section 6 names two unbuilt steps. They are not equal in cost, and conflating them would be a disservice to anyone deciding what to do next. So I ask the sharper question:

> *what is the* smallest *next step that converts a piece of the map into actual road—and how much road does it buy?*

My answer is Heilmann–Lieb via Godsil's *path-tree*, and I think it is a strikingly good bargain: a single classical construction that delivers both real-rootedness and the Ramanujan bound at once, with no interlacing machinery. Here is the plan, in enough detail to start.

The construction. Fix a vertex u of G. Godsil's path-tree $T(G, u)$ is the tree whose vertices are the paths in G starting at u, with one path joined to another when it extends it by a single edge (Figure 4). It is a finite forest, and it carries the whole proof on two facts.

The two facts that do the work.

(i) **Divisibility.** $\mu_G(x)$ divides $\mu_{T(G,u)}(x)$ (Godsil). The matching polynomial of the path-tree is a multiple of the one we care about, so any root bound for T transfers to G.

(ii) **Forests are gauge-trivial.** On a forest the matching polynomial *equals* the characteristic polynomial of the adjacency matrix: a tree has no even cycles to obstruct a global sign change, so signs can be gauged away. Hence $\mu_{T(G,u)}$ is the characteristic polynomial of a real symmetric matrix, therefore real-rooted; and the spectral radius of a tree of maximum degree Δ is at most $2\sqrt{\Delta - 1}$.

Put (i) and (ii) together and μ_G is real-rooted with all roots in $[-2\sqrt{\Delta-1}, 2\sqrt{\Delta-1}]$—Heilmann–Lieb, the Ramanujan threshold, in one stroke. This is why the path-tree is the right stone and not just *a* stone: it collapses two separate-looking analytic claims into one combinatorial construction plus standard spectral facts Mathlib already has.

What is already in hand, and what is missing. The matching polynomial and its deletion recurrence—the engine of every path-tree induction—are exactly what this paper formalizes, so the proposal does not start from zero. What Mathlib still lacks, and what a follow-up must build, is: the path-tree construction itself (as a `SimpleGraph` on a path type); the divisibility lemma (i), whose textbook proof is an induction on the deletion recurrence; and the forest identity (ii). I flag one concrete friction I already met and that this step will meet again: the matching-number recurrence has a boundary case (m_N of a graph after deleting two adjacent vertices vanishes) that needs a small isolated-vertices lemma to discharge cleanly—minor, but the kind of detail a formalization cannot wave away.

An open invitation. I state this as a proposal, not a result: none of (i), (ii), or the path-tree itself is yet formalized, and I do not claim them. But the route is classical, the prerequisites are now in place, and the payoff—the first machine-checked Heilmann–Lieb—is well defined. If this paper has a sequel, this is its first sentence.

8 Implications

What might this small piece of work be good for? I list three kinds of implication, in decreasing order of how confidently I can claim them, and a methodological aside.

Reusable, certified building blocks. The matching polynomial, its deletion recurrence, the signed adjacency matrix, and the 2-lift characteristic-polynomial factorization are now machine-checked and available to anyone working on matchings or signed-graph spectra in a proof assistant; Mathlib had none of them. Concretely, these are exactly the pieces a future formalization would build on: Heilmann–Lieb via Godsil's path-tree (Section 6) reuses the matching polynomial and its recurrence directly; the Gallai–Edmonds structure theorem and the independence polynomial sit on the same combinatorial infrastructure; and the signed-adjacency average is the precise recipe behind the broader *method of interlacing polynomials*—the very technique Marcus, Spielman and Srivastava used to settle the Kadison–Singer problem [7]. The $\mathbb{Z}/2$ sign-averaging lemmas are tiny, but the paper shows they are reusable across two levels (elementary-symmetric and power-sum), and I expect them to recur wherever an expected characteristic polynomial over ± 1 signs appears. This is the implication I can state most firmly: a handful of new, certified foundations.

A stepping stone toward formalized Ramanujan existence. The Marcus–Spielman–Srivastava theorem—Ramanujan graphs of every degree—is celebrated, and no proof assistant has yet touched it. This paper formalizes its structural half and maps the analytic half (Section 6) to scale. Should the landmark ever be built, these are foundations it can stand on, and the map may spare the next person the literature survey I had to run.

The mathematics, with a note of caution. Ramanujan graphs and expanders are not abstract curiosities: they underpin error-correcting codes (LDPC and expander codes), derandomization (Reingold's log-space algorithm for undirected connectivity uses expander walks), fast Laplacian solvers, cryptographic hash functions built from isogeny/expander graphs, and robust communication networks—that is why the underlying theorem matters. I am wary, though, of borrowing that importance for this paper. Formalizing a foundational identity does not by itself ship a code or a network; its value lies in certified foundations and in the growing corpus of formalized mathematics, not in immediate application. I claim the former and not the latter.

A methodological aside. This development was carried out with an AI system as a research instrument: it proposed definitions, lemmas and tactics, and a build-as-oracle loop verified or rejected each against the kernel. The careful accounting in this paper— every claim sized to exactly what was proved, the limitations stated as plainly as the results—is itself part of what I hope the work illustrates: that AI-assisted formalization can avoid the over-claiming the method invites, precisely because the proof assistant, not the assistant, has the last word.

9 Related work

The mathematics is classical: Godsil–Gutman [3], Heilmann–Lieb [5], the Marcus–Spielman–Srivastava program [6, 7], and the 2-lift / Bilu–Linial picture underlying it; the moment-level companion is recent [1]. On the formalization side, Mathlib [10] provides simple graphs and characteristic polynomials but neither the matching polynomial nor signed-graph spectral theory. The development sits in a growing line of Lean combinatorics formalizations—Hall's marriage theorem [4], the Kruskal–Katona theorem [8], Szemerédi's regularity lemma [2], and proof pearls in the same spirit [9]—none of which touches matching polynomials or signed-graph spectra. The interlacing toolkit of PerAlexandersson's `RealRooted` is a research codebase, not a polished library, and does not include the matching polynomial; my planned path-tree route (Section 7) sidesteps it. I am not aware of any prior machine-checked treatment of these results.

10 Conclusion

I set out from a simple question—what is the average characteristic polynomial of a random signing?—and followed it, in a proof assistant, to a complete and checked answer (Godsil–Gutman), to the small $\mathbb{Z}/2$ fact that makes it tick, to a second identity that shares that fact, and to the matrix bridge that points it all at Ramanujan graphs. None of these results is new mathematics; the contribution is that they are now machine-verified, that the matching polynomial exists in a prover for the first time, and that the shared engine is visible in the dependency graph rather than asserted. The deep half of the story—Heilmann–Lieb and interlacing families—remains ahead of me, mapped but unbuilt. I offer this as a pearl and an invitation.

Artifacts. All Lean sources, the figures of this paper, and the numerical cross-checks of Table 2 are available in the accompanying repository.

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